Cryptography - Module 2

Symmetric Key Algorithms

DES (Data Encryption Standard) uses a 56-bit key and operates on 64-bit blocks through 16 rounds of substitution and permutation. It is now considered insecure due to its small key size, vulnerable to brute-force attacks.

AES (Advanced Encryption Standard) replaced DES, supporting key sizes of 128, 192, or 256 bits over 10, 12, or 14 rounds respectively. AES operates on a 4x4 byte state matrix using SubBytes, ShiftRows, MixColumns, and AddRoundKey steps.

Cryptography - Module 2

RC4 Stream Cipher

RC4 generates a pseudorandom keystream that is XORed with plaintext. It uses a key-scheduling algorithm (KSA) to initialize a 256-byte state array, followed by a pseudo-random generation algorithm (PRGA) to produce output bytes one at a time. Known weaknesses in its keystream have led to it being deprecated in modern protocols.

Cryptography - Module 2

RSA (Asymmetric)

RSA relies on the difficulty of factoring large prime products. Key generation: choose primes p and q, compute n = p*q and phi(n) = (p-1)(q-1), pick public exponent e coprime to phi(n), compute private exponent d as the modular inverse of e mod phi(n). Encryption: c = m^e mod n. Decryption: m = c^d mod n.